

RecallIT — Flashcard Learning System

Report Overview

Project Name	RecallIT
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Abstract

This report presents the Data Flow Diagram (DFD) design of the RecallIT system, a flashcard-based learning platform with authentication, deck and card management, FSRS-based study scheduling, sketch-enabled content authoring, and analytics. Two levels of DFD are documented: DFD-0 (context-level view) and DFD-1 (decomposed process-level view). Together, these diagrams describe how data moves between users, core processes, and persistent data stores in the system.

Introduction

RecallIT is designed to support structured learning through spaced repetition. The system allows users to register/login, manage decks and cards, study due cards, submit review ratings, and analyze progress through a metrics board. DFD modeling is used to clarify the boundary of the system, major processes, data stores, and flow dependencies.

Objective of DFD Modeling

The objective of creating DFDs for RecallIT is to:

- Identify major data inputs and outputs of the system.
- Show interactions between user actions and internal processes.
- Define how data is stored and retrieved from system databases.
- Provide a clear reference for implementation, validation, and maintenance.

DFD-0: Context Diagram

DFD-0 provides the highest-level view of the system. At this level, RecallIT is treated as a single process interacting with one external entity (User) and multiple logical data stores.

Key Interpretation

- The **User** sends requests such as authentication, deck/card CRUD operations, study actions, and metrics queries.
- The system returns responses including auth status, deck/card payloads, next-card data, metrics summaries, and save confirmations.
- System persistence is distributed across Users, Decks, Cards, and Review Logs data stores.

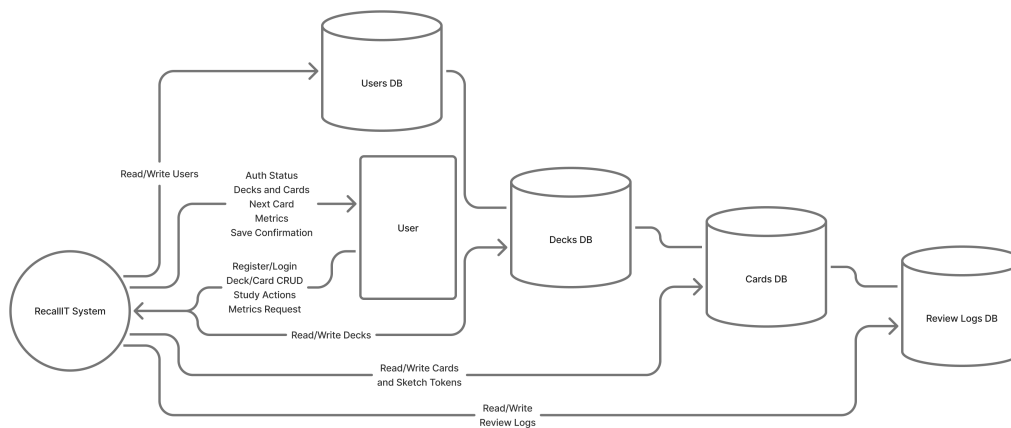


Figure 1: DFD-0 (Context Diagram) of RecallIT

DFD-1: Decomposition Diagram

DFD-1 breaks the single process into major functional modules and shows internal flow dependencies.

Main Processes

1. **Auth Management:** Handles registration/login, user verification, and token/auth response generation.
2. **Deck Management:** Supports deck creation/listing and deck-level ownership-scoped access.
3. **Card and Sketch Management:** Handles card CRUD, sketch token insertion, and card retrieval for authoring.
4. **Study Session and FSRS Scheduling:** Retrieves due cards, reveals answer

content, accepts rating feedback, and updates scheduling metadata.

5. **Metrics and Analytics:** Aggregates review logs and card/deck status into summary metrics and chart-ready analytics.

Data Stores and Roles

- **Users DB:** Stores identity and authentication-related user records.
- **Decks DB:** Stores user-owned deck metadata.
- **Cards DB:** Stores card content, sketch references, due timestamps, and FSRs state.
- **Review Logs DB:** Stores review events and rating history for analytics and progress tracking.

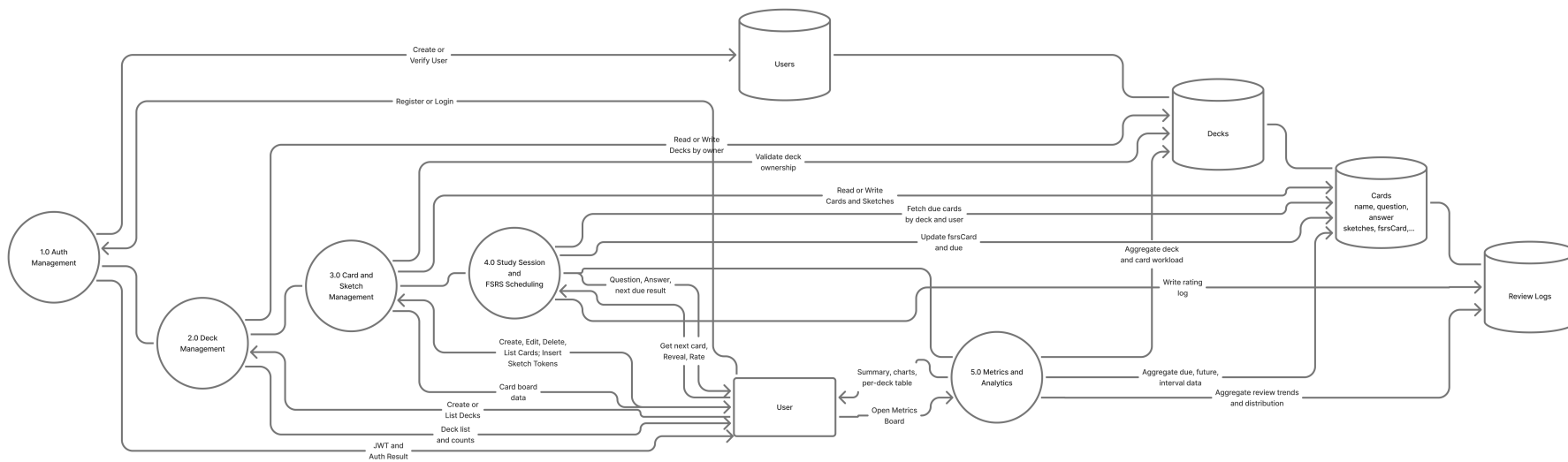


Figure 2: DFD-1 (Level-1 Decomposition) of RecallIT

Consistency Between DFD-0 and DFD-1

The decomposition is balanced:

- External interactions visible in DFD-0 are preserved and refined in DFD-1.
- Data stores in DFD-0 are mapped to process-level read/write behavior in DFD-1.
- User-facing outputs are traced to explicit generating processes.

Conclusion

The DFD model of RecallIT captures both macro-level and process-level data movement in a consistent way. DFD-0 defines system boundaries and major storage endpoints, while DFD-1 clarifies functional decomposition and data responsibilities. This report serves as a design artifact for implementation verification, team communication, and future extension planning.